

import

```
In [1]: sqrt(49)           # direct sqrt is not a function
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[1], line 1  
----> 1 sqrt(49)  
NameError: name 'sqrt' is not defined
```

import math

```
In [7]: import math  
math.sqrt(49)           # math is module then sqrt() is inbuilt function
```

```
Out[7]: 7.0
```

```
In [10]: x=2.3  
x
```

```
Out[10]: 2.3
```

```
In [11]: math.floor(2.3)           # 2 is floor ( floor means consider their lower value)
```

```
Out[11]: 2
```

```
In [12]: math.ceil( 5.4)           # 6 is ceil (ceil means consider their upper value)
```

```
Out[12]: 6
```

```
In [13]: math.floor(4.5)
```

```
Out[13]: 4
```

```
In [14]: math.ceil(6.8)
```

```
Out[14]: 7
```

```
In [15]: round(5.4)
```

```
Out[15]: 5
```

```
In [16]: round(4.3)
```

```
Out[16]: 4
```

```
In [18]: t.floor(4.4)           # m is not a function
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[18], line 1
----> 1 t.floor(4.4)      # m is not a function

NameError: name 't' is not defined
```

import math as

```
In [19]: import math as t
         t.floor(4.5)
```

```
Out[19]: 4
```

```
In [20]: t.ceil(6.8)
```

```
Out[20]: 7
```

```
In [21]: # all codes are run at a time that's calles code optimization
```

from math import *

```
In [28]: from math import *
         print(math.floor(5.4))
         print(math.ceil(5.7))
         print(math.pow(5,3))      # power & sqrt is in float
         print(math.sqrt(64))
```

```
5
6
125.0
8.0
```

```
In [29]: print(math.ceil(5.8))
```

```
6
```

input()

```
In [2]: x=input()    # bcoz in input whatever you put,it always consider as a string
         x
```

```
Out[2]: '5'
```

```
In [4]: x=input()      # 2 input => 2 console
         y=input()
         z=x+y          # str1 + str1 =str1str2
         print(z)
```

```
45
```

```
In [5]: print(type(x))
         print(type(y))
```

```
print(type(z))
```

```
<class 'str'>
<class 'str'>
<class 'str'>
```

```
In [ ]: # type casting is use for str to in
```

```
In [8]: f=int(input("enter a number:"))
        t=int(input("enter a number:"))    # using int we convert str into int
        print(f*t)
```

```
30
```

```
In [9]: print(type(f*t))
```

```
<class 'int'>
```

input with indexing

```
In [14]: f=input('enter a number:')[1:4]

        print(f)
```

```
483
```

```
In [15]: t=input('enter a string:')[3:]
        print(t)
```

```
lrani
```

```
In [22]: t=int(input('enter a number:'))[4:7]    # int is not used during indexing and
        print(t)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[22], line 1
----> 1 t=int(input(                ))[4:7]    # int is not used during indexi
ng and slicing
      2 print(t)

ValueError: invalid literal for int() with base 10: 't37823y22732'
```

```
In [1]: T=int(input('enter a number:'))[5]
        print(T)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[1], line 1
----> 1 T=int(input(                ))[5]
      2 print(T)

TypeError: 'int' object is not subscriptable
```

input function with eval for expression

```
In [3]: x=input('enter a string')    # this is expression not no.
        print(x)                     # it involves no. & operator
```

4+6-3

```
In [5]: y=int(input('enter a string          # in expression int is not allow
print(y)
```

```
Cell In[5], line 1
    y=int(input('enter a string          # in expression int is not allow
          ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
In [6]: y=eval(input('enter a string'))
print(y)
```

10

```
In [7]: w=eval(input('enter a string'))
print(w)
```

24

```
In [ ]:
```